

Software Verification and Validation Plan

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Section 1: Purpose

The purpose of this plan is to ensure that Verification and Validation guarantees that the product is correctly built and that it meets the needs of its users. The SVVP will be necessary when Verification and Validation is performed.

Section 2: Referenced Documents

Software Project Management Plan (SPMP)
Software Configuration Management Plan (SCMP)
Project requirements and design documents
IEEE 1012:2024 Standard
IEEE 829-2008 Example Test Scheme
ISO/IEC/IEEE 12207:2017 Standard

Section 3: Definitions

Verification: Checking that the product is built correctly according to the requirements.

Validation: Ensuring the product fulfills user needs and functions as intended.

V&V: Combined process of Verification and Validation.

Section 4: V&V Overview

4.1 Organization

The V&V process will be managed by our project manager and carried out by team members who are partly independent in development. The leader can oversee testing and documentation review, while developers handle code verification. These are coordinated with configuration management and project management through regular meetings and shared platforms (Github).

4.2 Master Schedule

The project will follow the life cycle of: Planning, requirements analysis, design, implementation, and testing. V&V tasks should occur every sprint to ensure that product is of quality. The feedback of V&V tasks should directly impact the product if needed.

4.3 Integrity Level Schema

The categories of the software integrity level scheme shown below are defined based on the seriousness of the

consequence(s) (of incorrect behavior during execution) and on the potential for mitigation(taking steps to lessen risk by lowering the probability of a risk event's occurrence or reducing its effect should it occur).

Level 4_Catastrophic

Level 3_Critical

Level 2_Marginal

Level 1_Negligible

4.5 Responsibilities

A person(everyone is still responsible for ensuring the product is of quality and meets customer expectations) other than the person who wrote the code is responsible for V&V tasks.

4.6 Tools, Techniques, and Methods

The V&V tasks will take place in VSCode and the browser that the website will be tested on is Google Chrome. V&V tasks will ensure that the product is functional and meets customer expectations.

Section 5: V&V Processes

5.3 Software V&V processes, activities, and tasks

System/Software Requirements Definition:

Verify that all backlog items (user stories) are Implemented. Check that each requirement is traceable and documented.

Design Definition:

Verify website structure, page layout, and navigation meet the design plan and ensure all assets (images, CSS, JS) are linked correctly.

Implementation:

Test website pages in Chrome for functionality, responsiveness, and broken links.

Peer code review: each team member tests someone else's code and final validation by running a full demo on localhost.

7.5 Standards, practices, and conventions

V&V tasks should not be performed on unfinished code and should only be done on code that is being submitted. V&V tasks should ensure that code is complete and is working as intended.

Code should not have any abnormalities.

Section 8: V&V test documentation requirements